Team Foundry

CLAY EARTH DATASET FOUNDRY · TRACK 1 · TEAM 46

An AI agent says it found 15 data centers. Can ADB trust that dataset?

> We built the system that answers that question **with evidence**, step by step and claim by claim.

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OUR IDEA

Judge the work, not just the answer

Record

Every model call and every tool call is captured as an OpenTelemetry span: what the agent saw, what it decided, what each tool returned.

Verify

Every claim must cite the tool result it came from. Code checks the value is really there, and that each site is inside the area that was asked for.

Judge

Three AI judges from three different companies score the parts code can't, each vote tied to the step it relied on.

Nothing is judged on the final answer alone. The trace is the evidence.

LIVE

Judging a dataset the agent already built

1. **The run:** "data centers within 10 miles of national or state parks in Texas", already built by Kimi K3 on AgentCore
2. **Judge again:** three judges start. Kimi steps down for its own agent; Mistral sits.
3. **While they think:** 15 / 15 claims verified against the tools, 8 / 8 sites inside Texas, on the map
4. **The verdicts land:** a score and how far to trust it
5. **Every vote points at evidence:** click a judge's reason, land on the tool call

▶ **Judge this run live**

Contrast: a run that fails

Backup: India solar parks

If time: a run an expert reviewed

The score moves a little each time the judges re-score; that's what the confidence label is for.

HOW THE SCORE WORKS

Code checks what code can. Judges assess the rest.

Measured by code

Provenance: is each value in the tool result it cites?

Area: is each site inside the boundary asked for?

+

Three AI judges

8 criteria, from tool choice to honest uncertainty.

Opus 5.5 · DeepSeek V3.2 · Kimi K3

No judge grades its own family's agent.

→

Median vote

Three votes, no ties: the median is always a real judge's vote.

→

Score + confidence

High · Medium · **Low**: how far the judges agree.

No answer: **FAILED**.

Judges never see the code's results, so code and judges are two independent lines of evidence.

DOES THE EVALUATOR ITSELF WORK?

We sabotaged runs on purpose. It noticed every time.

PLANTED DEFECT	CRITERION IT SHOULD HIT	BEFORE → AFTER
Searches with no area	Tool arguments	2 → 0
A made-up site	Grounding	2 → 0
The answer removed	Completeness	2 → 0

Live calibration, judges on a 0–2 scale. Its first pass caught a bug in our own evidence builder.

Judges agree like experts do

0.50

AGREEMENT SCORE

Human experts in the same kind of study: **0.56**.

Krippendorff's α: 1 = perfect, 0 = chance.
95% CI 0.12–0.67 over 7 runs.

A HUMAN IN THE LOOP

Experts correct the judges, and the judges learn

On one run: reconsider

Expert reviews → Judges re-check the trace → Score revised

The review is evidence to check, not an order. A judge may keep its vote, and says why. The original score is kept.

> "The steps show eight sites were checked against imagery, not three: five Central Valley sites in T6 and three North Coast sites in T9."
>
> Opus, correcting our deliberately wrong review (live, California vineyards)

Across runs: guidance

Reviews → Lessons → Held-out test → A person approves

Approved lessons reach every judge's prompt, under the rubric's wording. Each score records the version it used, and it can be rolled back.

> Stored in each deployment's own archive: not in the code, not in a browser.
>
> A third of reviewed runs are held out, so a lesson is tested on runs it never saw

Can a review push the judges? Live: ✓ WRONG REVIEW RESISTED ✓ "I INSIST" RESISTED ✓ INJECTED INSTRUCTIONS RESISTED

THE TRACES DON'T LIE

It caught bugs in our own agent

BEFORE

FAILED

Texas parks prompt: no answer, **$1.50** spent. The trace showed why, step by step.

AFTER THE FIXES

85.4 · High

Same prompt on AgentCore: 0 errors, 15 / 15 claims verified, 8 / 8 sites inside Texas, **$0.81**.

"Northern California" resolved to a 20-metre building in San Francisco, so every search came back empty.

LGND sign-in expired after exactly 300 seconds; every call after 290 s failed.

Narrowed searches (a 16 km circle around a park) were silently widened to all of Texas.

BUILT TO PLUG IN

Every part is swappable by config

Agent

Strands on Amazon Bedrock AgentCore; Claude Sonnet 5.5 or Kimi K3

Tools (MCP)

AWS Open Data (RODA) and LGND Geo imagery search

Trace

OpenTelemetry spans to an S3 archive; exporter swapped by config, proved in CI

Evaluate

Measured checks + judge tribunal on Bedrock; Team Foundry console (FastAPI)

ADB'S RUBRIC → ONE YAML FILE

JUDGES → CONFIG

MODELS → CONFIG

TRACING BACKEND → CONFIG

INFRA → CDK

Replays work offline: any archived run re-scores from its trace, without calling the agent again.

WHAT IT PROVES, AND WHAT'S NEXT

Honest about the limits

It proves today

Each claim matches the tool output it cites, each site is in the requested area, and three independent judges agree how well the work was done.

Not yet

Whether a site is **really** a data center. That needs an answer key, and ADB's own rubric.

Next

An answer key (e.g. US solar sites), ADB's rubric as a file, **ADB's experts reviewing runs** (the loop is built; the reviews are theirs to give), plain-language verdicts, and checking every condition in the prompt.

Plain-language verdicts and prompt compliance are being built today, as separate slices.

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Thank you. Questions?

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OPEN SOURCE · APACHE-2.0 github.com/onemorepereira/team46-breaking-barriers-hackathon